

Effect of thermal mass on life cycle primary energy balances of a concrete- and a wood-frame building

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ABSTRACT

In this study we analyze the effect of thermal mass on space heating energy use and life cycle primary energy balances of a concrete- and a wood-frame building. The analysis includes primary energy use during the production, operation, and end-of-life phases. Based on hour-by-hour dynamic modeling of heat flows in building mass configurations we calculate the energy saving benefits of thermal mass during the operation phase of the buildings. Our results indicate that the energy savings due to thermal mass is small and varies with the climatic location and energy efficiency levels of the buildings. A concrete-frame building has slightly lower space heating demand than a wood-frame alternative, due to the higher thermal mass of concrete-based materials. Still, a wood-frame building has a lower life cycle primary energy balance than a concrete-frame alternative. This is due primarily to the lower production primary energy use and greater bioenergy recovery benefits of the wood-frame buildings. These advantages outweigh the energy saving benefits of thermal mass. We conclude that the influence of thermal mass on space heating energy use for buildings located in Nordic climate is small and that wood-frame buildings with cogeneration based district heating would be an effective means of reducing primary energy use in the built environment.

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1. Introduction

Buildings are complex systems comprising multiple components and are dynamic due to their different life cycle phases and processes. Energy is used during the life cycle of buildings for material production, transport, construction, operation, maintenance and demolition. An understanding of the dynamics of these activities as well as their interactions is needed to design strategies to minimize the energy and climate impacts of buildings.

Studies of building systems show that the choice of building material affects the primary energy use and the greenhouse gas emissions of buildings. Cole and Kernan [1] analyzed the total life cycle energy use of a building constructed with wood, steel, or concrete structural material. They found that the concrete and steel buildings used more energy than the wood building. Cole [2] investigated the energy use and greenhouse gas emissions due to the on-site construction activities of buildings made with wood, steel or concrete structural material. He found that the energy used and greenhouse gas emissions were lowest for constructing the steel building, slightly higher for the wood building, and significantly higher for the concrete building. Adalberth [3] quantified

the primary energy use of functionally equivalent buildings with wood and concrete frames. She found that the operation energy was slightly lower for the concrete-frame building than for the wood-frame building, but the overall life cycle energy balance, including the production, operation and end-of-life phases was 1% lower for the wood-frame building than for the concrete-frame building. Gustavsson et al. [4] calculated the primary energy and CO₂ balances of buildings constructed with wood or concrete frames, taking into account various life cycle parameters including energy available from biomass residues from logging, wood processing, construction, and demolition. They found that the wood building used less energy and emitted significantly less CO₂ to the atmosphere than the concrete building. Gustavsson and Sathre [5] explored the variability in primary energy and CO₂ balances of wood and concrete buildings. They found that recovery of biomass residues has the single greatest effect on the primary energy and carbon balances of the buildings, followed by land use issues and concrete production parameters. Dodoo et al. [6] investigated the effects of post-use material management on life cycle carbon balances of buildings, and compared the carbon balance of a concrete- and a wood-framed building, taking into account various post-use material management options and carbonation during the service life and end-of-life phases of the buildings. They found that carbonation of crushed post-use concrete gives significant uptake of CO₂, but still a wood-frame building has a substantially lower life

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cycle carbon balance than a concrete-frame building. However, studies of the impact of thermal mass on the life cycle primary energy use are scarce.

Thermal mass describes the heat storage capacity of a material and it indicates the ability of the material to provide inertia against temperature variations. Effective thermal mass material can absorb and store significant amounts of heat, and this can help to level out temperature variations. The thermal mass of a material is a function of different properties including the heat capacity, density and thermal conductivity of the material. Materials with high heat capacity and density absorb and store considerable amounts of heat when exposed to an external heat source. Thermal conductivity influences the time lag of absorbing or releasing heat. If the thermal conductivity of the material is too high the stored heat is released quickly when the heat source is removed. The material mass of buildings can be designed to store the internal and solar heat gains to moderate temperature swings and thereby reduce the energy use for space conditioning. The effectiveness of thermal mass in buildings depends on the interactions of several parameters including climatic location, orientation, window area, insulation, ventilation, load profile and occupancy pattern of buildings [7,8]. Hence, designing to optimize thermal mass of buildings is a complex issue, and requires careful analysis [9]. Thermal mass is more effective in climates where the diurnal range of temperature is high. Insulation generally tends to interact with thermal mass, with the benefit of thermal mass decreasing with increased insulation [7]. Combinations of construction techniques, load leveling and plant operation scheduling can be used to optimize the thermal mass of buildings [10].

Several studies have investigated the effects of thermal mass on the energy use of buildings in different climates. Kalogirou et al. [11] investigated the effects of thermal mass on the heating and cooling loads of buildings in Cyprus, where the diurnal temperature variation is large and hence favorable for thermal mass. They found that thermal mass substantially reduced the heating load and slightly increased the cooling load of buildings. However, Aste et al. [12] explored the effect of thermal mass in an Italian building by analyzing the energy performance of wall systems with the same U -values but different thermophysical properties including heat transfer surface, specific heat capacity, thermal conductivity and solar control factor. They found that the walls which performed best in terms of energy had effective combination of dynamic thermophysical properties, mainly greater heat transfer surface, and not necessarily the best thermophysical properties. Thus the heat transfer surface influences the effectiveness of thermal mass for saving energy, and careful design and control strategies are required for effective use of thermal mass. Gregory et al. [13] presented a comparative analysis on the influence of thermal mass in simplified building modules in Australia. They found that thermal mass reduced the space heat demand significantly, particularly where the thermal mass was placed at the interior side of the insulation. Zhu et al. [14] compared identical buildings constructed with wood and concrete frames in a hot US climate where thermal mass is considered favorable. They found that the wood-frame building used more space heating energy, but less space cooling energy than the concrete-frame building. Kalema et al. [15] varied the frame material of a case-study building between extra-lightweight to heavyweight, and studied the effect of thermal mass on the final energy for space heating and cooling buildings in the Nordic climate. They calculated the final energy savings of thermal mass to be about 4–16%. However, Josikalo and Kurnitski [16] employed a similar approach and found that the heavyweight building used 0.7% less space heating than the lightweight building, and concluded that the effect of thermal mass is insignificant for a Finnish apartment building. However, the above studies mostly focused on final energy use for space heating

during the operation phase of buildings, and did not include other building life cycle activities.

The operation energy generally dominates the life cycle energy use of buildings so minimizing space heating demand is important. However, focusing only on optimizing the energy performance in the operation phase may result in potential trade-offs in other life cycle phases [17]. The significance of the other building life cycle phases and activities increases as measures are implemented to reduce space heating demand [18]. The various processes along the energy supply chain, from the extraction of raw material to refining, transport, conversion to heat and electricity, and distribution to the user can be performed with different energy efficiencies.

A comprehensive analysis of the impacts of thermal mass in buildings needs to include the various building life cycle activities and the full energy chains. Hacker et al. [19] investigated the impact of thermal mass and climate change on CO₂ emissions of a UK building when using different frame materials. They considered the emissions from the production and operation phases but did not quantify the emissions from the end-of-life phase of the buildings. In a series of reports, Marceau and Van Geem [20–22] presented results of life cycle assessments of concrete-based and wood-based buildings, including energy and emissions during the extraction of raw materials, construction, maintenance, and operation of the buildings. They reported the primary energy and environmental impacts of concrete-based buildings to be comparable to or lower than wood-based building, due to the thermal mass effect. However, the buildings they analyzed had different U -values and were not functionally equivalent. This makes it difficult to distinguish the effect of the different U -values and the effect of thermal mass due to the choice of material.

In this study, we explore how thermal mass influences the life cycle primary energy balances of functionally equivalent concrete- and wood-frame buildings. We vary the energy efficiency and location of the buildings as well as the energy supply system.

2. Methods

Our study is based on system analysis methodology in a life cycle perspective and we consider the entire energy chains and all life cycle phases, including the potential benefits associated with thermal mass of building material systems during the operation phase of buildings. The life cycle activities and primary energy flows included in the study are shown in Fig. 1.

2.1. General approach

We analyze functionally equivalent versions of a reference building with either a concrete or wood frame to explore the effect of thermal mass on the life cycle primary energy use of the buildings. The reference building is a 4-storey residential building with 16 apartments and a total heated floor area of 1190 m², located in Växjö in southern Sweden. Detailed information, including drawings and thermal properties, of the versions of the building with a concrete frame and a wood frame is reported by Persson [23]. Calculations were made of the mass of materials required to construct the concrete and wood versions of the buildings [3]. For the wood-frame building, the outer walls of the building consist of three layers, including plaster-compatible mineral wool panels, timber studs with mineral wool between the studs, and a wiring and plumbing installation layer consisting of timber studs and mineral wool. Two-thirds of the outer façade is plastered with stucco, with the remainder covered with wood paneling. The roof consists of two layers of asphalt-impregnated felt, wood panels, mineral wool between wooden roof trusses, polythene foils and plasterboards. The ground floor consists of oak boarding on

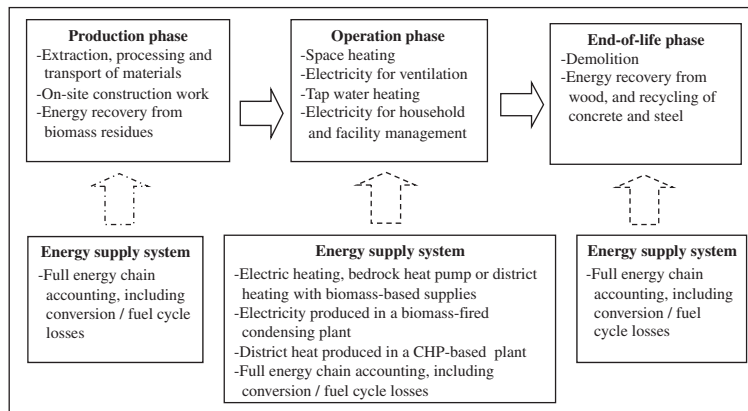


Fig. 1. Life cycle activities and primary energy flows modeled.

concrete slab laid on expanded polystyrene and crushed stone, and the remaining floors are made of light timber joists. For the concrete-frame building, reinforced concrete replaces the light timber joists floors, the load-bearing timber studs and walls, and the stucco and the plaster-compatible mineral wool panels of the external wall for the wood-frame version. Mineral wool insulation is fixed between the reinforced concrete walls and the outer façade of cement render. Fig. 2 shows wall and floor construction details for the functionally equivalent versions of the reference building with concrete and wood frames.

We use the original concrete- and wood-frame buildings as reference, to which we model changes including improved insulation, windows, doors, airtightness, and addition of ventilation heat recovery systems and efficient water taps, to achieve new buildings that meet the energy efficiency requirements of the current Swedish building code or the passive house standard [24,25]. Dodoo et al. [26] discussed and analyzed the life cycle primary energy implications of these standards. We modeled the energy use of the buildings to these standards to explore the role of thermal mass in buildings of varying levels of energy efficiency. Table 1 summarizes the thermal properties of all the buildings. Except for thermal mass, all the thermal properties are the same for the wood and the concrete versions of the buildings built to each energy efficiency standard.

2.2. Thermal dynamics of building envelope details

We model the envelope thermal properties of the buildings using the VIP+ simulation program by Strusoft [27]. The VIP+ is a dynamic energy balance program that models the hourly final energy balance of buildings taking into account several parameters

including envelope thermal properties, orientation, glass area, heating and ventilation systems, heat gains from lighting, appliances, human bodies and solar radiation, and operation schedule, indoor temperature, geographical location and outdoor climate. The software has been validated by the International Energy Agency building energy simulation test and diagnostic method [27,28]. VIP+ calculates the hourly variation of heat flux of buildings in detail, allowing for analysis of the role of thermal mass in reducing energy use. The program allows for one-, two- and three-dimensional heat flow modeling of various building envelope details, from a comprehensive materials and components catalogue, and calculates the *U*-values of the resulting building envelope configurations. VIP+ analyzes heat storage capacity of buildings taking into account various thermophysical properties including thermal conductivity, density and heat capacity of materials. It models heat storage of buildings assuming the configurations of the building envelope as a series of thermal resistance and capacitance with finite difference in response to thermal variations. This approach captures the dynamic heat exchange between the building envelope, processes and the environment, and is described in detail by Johannesson [29].

The external wall configurations may have strong influence on the thermal mass as these vary for the functionally equivalent concrete- and wood-frame buildings. To fully account for all construction details, we use the three-dimensional modeling to simulate the construction details of all the external walls. We use two-dimensional modeling for the roof and one-dimensional modeling for the ground floor, as the construction details of the ground floor and roofs are the same for the functionally equivalent versions of the buildings. For the doors and windows details, we apply pre-defined parameters from the VIP+ catalogue of building parts.

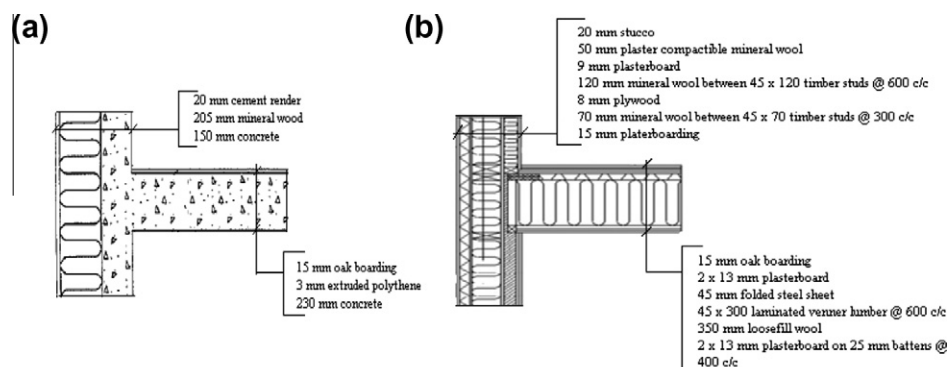


Fig. 2. Wall and floor construction details for the functionally equivalent (a) concrete-frame and (b) wood-frame reference building.

Table 1

Thermal properties for concrete- and wood-frame buildings designed to different energy standards.

Concrete-/wood-frame building	U-value (W/m ² K)					Air leakage at 50 Pa (l/s m ²)	Water taps	Mechanical ventilation
	Ground floor	External walls	Windows	Doors	Roof			
Reference	0.23	0.20	1.90	1.19	0.13	0.8	Conventional	Exhaust air
Current code	0.23	0.12	1.20	1.19	0.08	0.6	Efficient	Heat recovery
Passive standard	0.23	0.11	0.90	0.90	0.06	0.3	Efficient	Heat recovery

2.3. Production phase

We use a bottom-up approach to quantify the production primary energy balance of the buildings, including the entire energy chains and material losses during production. We calculate the production primary energy balance of the buildings as the primary energy expended to extract, process, transport and assemble the materials, minus the net energy of biomass by-products that can be recovered and made available for external use throughout the material life cycle [4]. The primary energy use to extract, process and transport the materials is calculated, based on Sathre [30], as:

$$E_{\text{production}} = \sum_i \left\{ \sum_k [F_{i,k} \times (1 + \alpha_k)] + \frac{L_i}{\eta} + B_i \right\}$$

where $E_{\text{production}}$ is the total primary energy use for material production (kW h); i , the individual types of materials in the building; F , the end-use fossil fuel energy used to extract, process and transport the materials (kW h); k , the fossil fuels: coal, oil, and natural gas; α the fuel cycle energy requirement of the fossil fuel; L , the end-use electricity to extract, process, and transport the materials (kW h); η , the conversion efficiency for electricity production; and B is the heat content (lower heating value) of the biofuels used in material processing (kW h).

We take account of the fuel cycle energy inputs, including extraction, transport, processing, conversion and distribution of the energy carriers. We assume this energy input to be 10% for coal, and 5% for oil and natural gas, of the delivered fuel [4]. The end-use electricity to produce the materials is assumed to be produced from a coal-fired plant with a 40% conversion efficiency and 2% distribution loss [5]. The embodied energy in the infrastructure used for the production, distribution and end-use of electricity and heat is expected to be very minor and is excluded.

Our data for the specific fossil fuel and electricity used to extract, process and transport the materials is based primarily on a Swedish study by Björklund and Tillman [31]. We also use specific energy use data from closely related studies [32–34] where data are not available from Björklund and Tillman [31]. Table 2 shows the specific end-use energy of selected materials.

The net energy available from biomass by-products is calculated based on Sathre [30], as:

$$E_{\text{byproducts}} = \sum_j \{M_j \times H_j \times [1 - \beta_j \times (1 + \alpha_{\text{diesel}})]\}$$

Table 2Specific end-use energy (kW h_{end use}/kg) to extract, process, and transport selected materials.

Material	Coal	Oil	Fossil Gas	Biofuel	Electricity
Concrete	0.09	0.10	–	–	0.02
Plasterboard	–	0.79	–	–	0.16
Lumber	–	0.15	–	0.70	0.14
Particleboard	–	0.39	–	1.40	0.42
Plywood	–	0.75	–	2.65	0.48
Steel (ore-based)	3.92	0.86	1.34	–	0.91
Insulation	2.00	0.36	0.02	–	0.39

where $E_{\text{byproducts}}$ is the net energy from recovered biomass residues (kW h); j , the different types of residues: forest, processing and construction; M , the mass of the recovered residue (oven dry tons); H , the lower heating value of the biomass residue (kW h/oven dry ton); β , the diesel fuel energy required to recover and transport the residue, expressed as a proportion of the heat energy contained in the residue; and α is the fuel cycle energy requirement of the diesel fuel.

We quantify the mass of the forest residue available from final fellings using biomass expansion factors from Lehtonen et al. [35]. We assume the recovery of 70% of available forest harvest residues and 100% of processing and construction residues. The diesel fuel used to recover the forest biomass residues is assumed to be 5% of the lower heat content of the recovered wood [4]. The primary energy for on-site assembly of the building materials is taken to be 50 kW h/m² for the reference wood-frame building and 100 kW h/m² for the reference concrete-frame building, based on Adalberth [3]. We assume that half of the energy used, counted as primary energy use, to assemble the material is electricity, and the other half is diesel fuel. We assume that the primary energy use to assemble the materials for the buildings with improved energy efficiency is proportionally equal to the reference buildings, weighted by the relative amounts of primary energy for material production.

2.4. Operation phase

We calculate the operation final energy use for space heating, ventilation, domestic hot water heating and household electricity with the VIP+ simulation program [27]. We assume an indoor temperature of 22 °C in the living areas and 18 °C in the common areas of the apartment building. We model the space heating demand of the buildings for three locations in Sweden: Växjö, Östersund and Kiruna, to explore the impact of climatic differences on the thermal mass and energy performance of the buildings. Table 3 shows the climatic data for the different locations.

Based on the operation final energy use, we calculate the operation primary energy use with the ENSYST software [36]. This software estimates primary energy use taking into account the entire energy chain from natural resources extraction to supply of final energy. We use the software's default assumptions regarding the source, production and transport of primary fuels. We consider three different technologies for heating the buildings: electric resistance heating, bedrock heat pump, and district heating. An annual coefficient of performance of 3 for heating is assumed for the bedrock heat pump. For the electric resistance heating and the heat pump, 95% of the electricity is assumed to be supplied from stand-alone plant using biomass steam turbine (BST) technology and the remaining from light-oil gas turbine. We assume that 85% of the district heat is supplied from a combined heat and power (CHP) plant using BST technology, and light-oil boilers supply the remainder [37]. Dodoo et al. [38] provides a summary of key data input used in the ENSYST software for the primary energy analysis. We allocated the cogenerated electricity using the subtraction method, assuming that the cogenerated power replaces electricity

Table 3

Climatic data of the locations for thermal performance analysis of the buildings.

Location	Relative location	Latitude	Longitude	Annual outdoor temperature (°C)		
				Maximum	Minimum	Average
Växjö	South	56°9'N	14°5'E	28.0	−18.0	6.5
Östersund	Middle	63°2'N	14°4'E	24.0	−28.0	2.5
Kiruna	North	67°8'N	20°0'E	24.0	−33.0	−1.2

that would instead have been produced in a stand-alone plant using the same fuel and technology as the CHP plant [39].

2.5. End-of-life phase

In the end-of-life analysis, we assume that the buildings are demolished by selective dismantling at the end of their service life, and then the concrete, steel and wooden materials are recovered. The end-of-life primary energy use is calculated as the net primary energy used to disassemble, transport and recover the building materials, minus the energy benefits from recycling and recovering the end-of-life materials [6,40]. The primary energy to demolish the reference building is taken to be 5 and 10 kW h/m² for the wood-frame and concrete-frame buildings, respectively [3]. We assume that the demolition primary energy use for the buildings with improved energy efficiency is proportionally equal to the demolition primary energy use for the reference building of similar frame, weighted by the relative amounts of material production primary energy use.

We follow the method used by Dodoo et al. [6] to calculate the primary energy implications of recovered concrete, steel and wood and assume that 90% of each material is recovered. The primary energy benefit of the end-of-life of these materials is calculated as the primary energy use avoided due to the recovered materials minus the primary energy used to recover the materials. All data for these calculations are given in Dodoo et al. [6]. The concrete is assumed to be crushed into aggregate, replacing crushed stones for below-ground applications. The steel is assumed to be recycled into feed-

stock, replacing either ore- or scrap-based steel. We assume that the demolished wood material is used for energy purposes as this is common in Sweden.

2.6. Complete lifecycle

We calculate the net primary energy use over the complete life cycle of the buildings, including the production, operation and end-of-life phases. We assume that the buildings have a 50-year lifespan. In a sensitivity analysis we vary various parameters, including the lifespan and energy supply system of the buildings to explore their overall significance on the life cycle primary use of the buildings.

3. Results

The mass of principal materials required for the construction of the buildings, and the primary energy for production of these materials, are presented in Table 4. Significant differences between the building mass of the concrete- and wood-frame buildings include insulation and plasterboard, besides the structural frame materials of concrete, steel and lumber. The largest contributors to the total building mass are concrete and crushed stone, together accounting for about 90% and 65% of the total building mass for the concrete- and wood-frame buildings, respectively. The average primary energy used for the production of these two materials, in relation to the total production energy of the buildings, is about 38% for the concrete-frame buildings and 9% for the wood-frame

Table 4

Mass of principal materials used in the buildings, and the primary energy for the materials' production.

Material	Mass (kg of air-dry material/m ²)						Primary energy use (kW h/m ²)					
	Reference		Current code		Passive standard		Reference		Current code		Passive standard	
	Concrete frame	Wood frame	Concrete frame	Wood frame	Concrete frame	Wood frame	Concrete frame	Wood frame	Concrete frame	Wood frame	Concrete frame	Wood frame
Concrete	1136	187	1142	193	1144	195	292	48	293	50	293	50
Iron/steel	21	13	21	13	21	13	187	116	187	116	187	116
Lumber	28	50	28	57	28	59	34	61	34	70	34	72
Particle board	15	15	15	15	15	15	42	44	42	43	42	43
Plywood	17	18	17	18	17	18	78	84	78	84	78	84
Insulation	8	18	19	28	20	31	29	62	67	100	70	110
Plasterboard	21	75	21	75	21	76	27	96	27	97	27	97
Crushed stone	265	265	270	270	271	271	6	6	6	6	6	6
Mortar	19	21	20	21	20	21	7	7	7	7	7	7
Blocks	3	3	3	3	3	3	3	3	3	3	3	3
Glass	3	3	5	5	5	5	9	9	13	13	13	13
Paper	2	2	2	2	2	2	16	16	16	16	16	16
Paint	1	1	1	1	1	1	5	5	7	7	7	7
Putty/fillers	3	3	3	3	3	3	8	8	8	8	8	8
White goods	2	2	2	2	2	2	8	8	8	8	8	8
Porcelain	1	1	1	1	1	1	3	3	3	3	3	3
Copper/zinc	1	1	1	1	1	1	3	3	3	3	3	3
Krypton	0	0	<1	<1	<1	<1	0	0	<1	<1	<1	<1
Low-coating	0	0	<1	<1	<1	<1	0	0	<1	<1	<1	<1

buildings. The contribution of crushed stone, used for the foundations, to the material production primary energy is very minor despite its significant mass.

The primary energy balance during the production phase of the buildings is summarized in Table 5. The negative numbers denote energy that is available from recovered biomass residues. The material production primary energy use is dominated by fossil fuels and electricity while the use of bioenergy is about 10–18%. The biomass residues for external use are about 23–56% of the total production energy. The wood-frame buildings have lower primary energy use for material production and on-site construction, and greater biomass residues for external use. The production primary energy balance of the concrete-frame buildings is at least twice that of the wood-frame alternatives.

The annual final energy use for space heating the buildings for the different cases is shown in Table 6. The annual profiles of final space heating demand of the concrete- and the wood-frame buildings arranged in descending order are shown in Figs. 3 and 4. Fig. 3 shows the profiles for all the buildings with location in Växjö, and Fig. 4 shows the profiles for the reference buildings in the different locations. The dashed lines show the concrete-frame buildings and the corresponding solid lines show the wood-frame alternatives. The energy savings of thermal mass is the difference between the dashed line and the corresponding solid line. The savings are insignificant during the winter season and increase somewhat during the spring and autumn seasons. The overall effect of thermal mass is annually about 0.3–0.7 kW h/m². The lower values are for more energy-efficient buildings.

Table 7 shows the annual primary energy use for space heating the concrete- and wood-frame buildings in three locations, with different end-use heating systems. The energy supply is based on BST technology. The absolute primary energy savings due to thermal mass is higher when using less energy efficient supply systems.

Table 8 shows the annual operation primary energy use for the buildings, with location in Växjö. The concrete-frame buildings

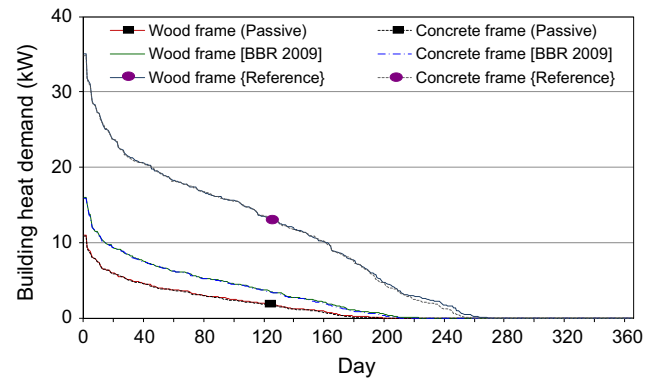


Fig. 3. Annual profiles of final space heating demand of the buildings located in Växjö arranged in descending order.

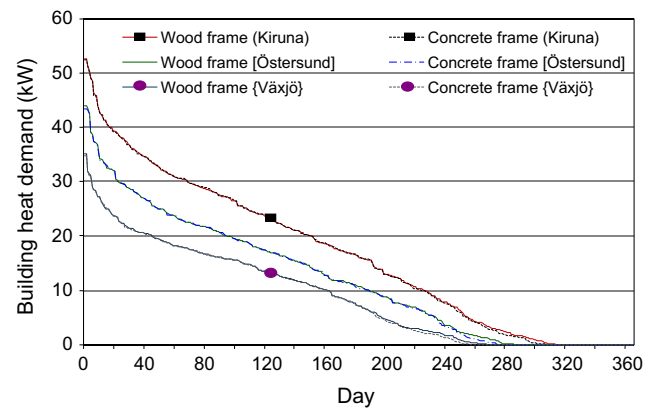


Fig. 4. Annual profiles of final space heating demand arranged in descending order for the reference buildings in the different locations.

Table 5

Primary energy balance (kW h/m²) during the production phase of the buildings.

Description	Reference		Current code		Passive standard	
	Concrete frame	Wood frame	Concrete frame	Wood frame	Concrete frame	Wood frame
<i>Material production energy use</i>						
Fossil fuels	474	306	507	346	510	353
Electricity	199	169	212	190	213	195
Bioenergy	84	103	84	108	84	108
Total	757	579	803	644	806	656
<i>On-site construction energy use</i>						
Fossil fuel	50	25	53	29	53	29
Electricity	50	25	53	28	54	29
Total	100	50	106	57	107	58
Total production energy use	857	629	909	701	913	714
<i>Biomass residue recovery</i>						
Forest harvest	–80	–125	–82	–139	–82	–143
Wood processing	–128	–220	–129	–251	–129	–258
Total	–208	–345	–211	–389	–211	–400
Production energy balance	649	284	698	312	702	314

Table 6

Annual final energy use for space heating (kW h/m²) of the buildings at different locations.

Location	Reference			Current code			Passive standard		
	Concrete frame	Wood frame	Difference	Concrete frame	Wood frame	Difference	Concrete frame	Wood frame	Difference
Växjö	68.30	68.95	0.65	23.92	24.32	0.40	14.73	15.08	0.35
Östersund	93.65	94.27	0.62	37.66	38.16	0.50	24.51	24.95	0.44
Kiruna	129.43	130.08	0.65	57.77	58.12	0.35	39.74	40.15	0.41

Table 7
Annual operation primary energy use (kW h/m²) for space heating for buildings in three locations, with different end-use heating systems and energy supply based on BST technology.

Description	Reference		Current code		Passive standard	
	Concrete frame	Wood frame	Concrete frame	Wood frame	Concrete frame	Wood frame
<i>Electric heated</i>						
Växjö	203.9	205.9	71.4	72.6	44.0	45.0
Östersund	279.6	281.5	112.4	113.9	73.2	74.5
Kiruna	386.4	388.4	172.5	173.5	118.7	119.9
<i>Heat pump heated</i>						
Växjö	76.1	76.8	26.7	27.1	16.4	16.8
Östersund	104.4	105.0	42.0	42.5	27.3	27.8
Kiruna	144.2	144.9	64.4	64.8	44.3	44.7
<i>District heated</i>						
Växjö	45.4	45.8	15.9	16.2	9.8	10.0
Östersund	62.2	62.6	25.0	25.3	16.3	16.6
Kiruna	86.0	86.4	38.4	38.6	26.4	26.7

Table 8
Annual primary energy use for operation (kW h/m²) for buildings located in Växjö.

Description	Reference		Current code		Passive standard	
	Concrete frame	Wood frame	Concrete frame	Wood frame	Concrete frame	Wood frame
<i>Electric resistance heated</i>						
Space heating	203.9	205.9	71.4	72.6	44.0	45.0
Ventilation electricity	12.1	12.1	25.4	25.4	25.4	25.4
Tap water heating	74.5	74.5	44.7	44.7	44.7	44.7
Household electricity	133.7	133.7	133.7	133.7	133.7	133.7
Total from operation	424.2	426.2	275.2	276.4	247.8	248.8
<i>Heat pump heated</i>						
Space heating	76.1	76.8	26.7	27.1	16.4	16.8
Ventilation electricity	12.1	12.1	25.4	25.4	25.4	25.4
Tap water heating	28.1	28.1	16.8	16.8	16.8	16.8
Household electricity	133.7	133.7	133.7	133.7	133.7	133.7
Total from operation	249.9	250.7	202.6	203.1	192.4	192.8
<i>District heated</i>						
Space heating	45.4	45.8	15.9	16.2	9.8	10.0
Ventilation electricity	12.1	12.1	25.4	25.4	25.4	25.4
Tap water heating	16.6	16.6	9.9	9.9	9.9	9.9
Household electricity	133.7	133.7	133.7	133.7	133.7	133.7
Total from operation	207.7	208.1	184.9	185.2	178.8	179.1

Table 9
End-of-life primary energy balances (kW h/m²) for the buildings.

Description	Reference		Current code		Passive standard	
	Concrete frame	Wood frame	Concrete frame	Wood frame	Concrete frame	Wood frame
Demolition energy use	10	5	11	6	11	6
Concrete recycling	−19	−3	−19	−3	−19	−3
Steel recycling	−96	−60	−96	−60	−96	−60
Wood recovery for bioenergy	−214	−305	−214	−333	−214	−340
Energy balance	−319	−363	−318	−390	−318	−397

have slightly (<1%) lower operation primary energy use than the wood-frame alternatives, due to the benefits of thermal mass. The primary energy use for tap water heating and household electricity constitute a significant part of the operation primary energy.

Table 9 shows the primary energy balance for the end-of-life phase of the buildings. The wood-frame buildings give greater end-of-life primary energy benefit than the concrete alternatives. Energy recovery from demolition wood gives large primary energy benefit, and less benefit is achieved through recycling the steel and concrete.

Table 10 shows the total life cycle primary energy use of the concrete- and wood-frame buildings in Växjö, with an assumed

building lifespan of 50 years. The end-use heating is based on district heating and the energy supply is based on BST technology. The lower production primary energy use and greater end-of-life benefits of the wood buildings more than offset its additional operation primary energy use, and this gives the wood-frame building 4–5% lower net life cycle primary energy balance than the concrete-frame alternatives.

4. Sensitivity analysis

We show the impacts of variation of several major parameters on the primary energy use of the buildings. We vary: the lifespan

Table 10Life cycle primary energy use and balance (kW h/m²) for district-heated buildings in Växjö, with an assumed life time of 50 years. The energy supply is based on BST technology.

Description	Reference		Current code		Passive standard	
	Concrete frame	Wood frame	Concrete frame	Wood frame	Concrete frame	Wood frame
<i>Production energy use</i>						
Material production	757	579	803	644	806	656
On-site construction	100	50	106	57	107	58
Total	857	629	909	701	913	714
<i>Operation energy use</i>						
Space heating	2268	2290	794	808	489	501
Ventilation electricity	605	605	1272	1272	1272	1272
Tap water heating	829	829	497	497	497	497
Household electricity	6684	6684	6684	6684	6684	6684
Total	10,386	10,407	9247	9261	8942	8954
<i>End-of-life energy use</i>						
Demolition	10	5	11	6	11	6
Total energy use over life cycle	11,253	11,041	10,167	9968	9866	9674
Production biomass residues	−208	−345	−211	−389	−211	−400
<i>End-of-life recycling and recovery</i>						
Concrete recycling	−19	−3	−19	−3	−19	−3
Steel recycling	−96	−60	−96	−60	−96	−60
Wood recovery for bioenergy	−214	−305	−214	−333	−214	−340
Total	−537	−713	−540	−785	−540	−803
Total energy balance over life cycle	10,716	10,328	9627	9183	9326	8871

of the buildings from 50 to 100 years; the energy supply technology from BST to biomass integrated gasification combine cycle (BIGCC); and recycled steel substitution from 100% ore-based to 50% ore-based and 50% scrap steel. We also consider the potential truncation error resulting from a bottom-up analytical approach. Our analysis is based on a bottom-up approach and gives detailed information about the individual processes and systems studied. This approach facilitates comparison of alternative buildings systems, for which a top-down approach is less suited because it uses aggregate data and thus integrates all flows in an entire sector. Some top-down studies assert that a significant share of the energy use in the production phase is indirect and is not recognized when using the bottom-up approach, resulting in truncation energy outside of the system boundaries. For example, Nässén et al. [41] estimated the truncation energy for material production when applying bottom-up instead of top-down approach to be about 21%. The authors, however, noted that there are several uncertainties related to the top-down approach and this could result in uncertainty in their estimate. Nevertheless, we show the impact of such truncation energy on the production primary energy use of the buildings.

The results of the sensitivity analysis in Table 11 indicate that the wood-frame buildings have consistently lower net life cycle primary energy balance than the concrete-frame alternatives. Including the truncated primary energy for material production has a small effect on the life cycle primary energy balances of the buildings. Changing the lifespan of the buildings from 50 to 100 years approximately doubles the primary energy balance of the buildings. This also reduces the differences in primary energy balance between the concrete- and wood-frame buildings by 3–6% compared to the base case. This is because the thermal mass benefits of concrete accumulate as the lifespan of the buildings gets longer, increasing the operational energy advantage of the concrete-frame building. Using BIGCC instead of BST reduces the primary energy balances of the buildings, and also reduces the operation primary energy advantage of the concrete-frame buildings. This is because with a more efficient energy system, the primary energy benefits of the thermal mass are less significant.

5. Discussion and conclusions

Total primary energy use largely determines environmental impact of material production and energy supply activities [42]. In this study we analyze the effect of thermal mass on space heating energy use and life cycle primary energy balances of a concrete- and a wood-frame building. Our analysis is based on functionally equivalent buildings and involves detailed hour-by-hour dynamic modeling of the building mass configurations, including the interaction between the different layers of the building mass configurations.

Our results indicate that a concrete-frame building has slightly lower space heating demand than a wood-frame alternative, due to the benefit of thermal mass in the concrete-based system. The energy savings benefit due to thermal mass varies with the climatic location and energy efficiency levels of the buildings, confirming the findings of Balaras [7]. The difference between the space heating demands of the alternative buildings is insignificant during the winter season and increase somewhat during the spring and autumn seasons. Overall, the final energy savings benefit due to thermal mass are small both in absolute terms (0.3–0.7 kW h/m²) and in relation to the final space heating demand (0.5–2.4%). These results are comparable to Josikalo and Kurnitski [16] and Ståhl [43], who found space heating energy savings benefit of thermal mass to be around 0.7–2% for buildings in the Nordic climate. Norén et al. [44] analyzed the effect of thermal mass in Swedish buildings and concluded that the benefit of thermal mass is small where buildings located in a Nordic climate have ample insulation and plasterboard cladding. However, Kalema et al. [15] found the space heating energy reduction due to thermal mass to range from 4% to 16%, for Nordic buildings. They used a quasi-steady approach to estimate the heat capacity and time constant associated with the building mass, which is an unsuitable approach because the interaction between building mass configuration and thermal condition is complex and detailed dynamic analysis is needed to accurately determine the impact of thermal mass.

The life cycle primary energy balance of the studied wood-frame building was lower than that of the concrete-frame alternative, due primarily to the lower production primary energy use and

Table 11
Impacts of varying selected parameters on the primary energy balances of the buildings with a life time of 50 years if not stated otherwise.

Parameter	Base case (BC)	Variations	Primary energy use (kW h/m ²)											
			Reference				Current code				Passive standard			
			Concrete frame (CF)	Wood frame (WF)	Difference (CF–WF)	Change from reference difference	Concrete frame (CF)	Wood frame (WF)	Difference (CF–WF)	Change from reference difference	Concrete frame (CF)	Wood frame (WF)	Difference (CF–WF)	Change from reference difference
<i>Production energy use</i>														
Reference	–	–	857	629	228	0	909	701	208	0	913	714	199	0
Truncated material energy	–	+21%	1037	761	276	48	1100	848	252	44	1105	864	241	42
<i>Operation energy use</i>														
Reference	–	–	10,386	10,407	–21	0	9247	9261	–14	0	8942	8954	–12	0
Building lifespan	50 years	100 years	20,772	20,814	–42	–21	18,494	18,522	–28	–14	17,884	17,908	–24	–12
Energy supply	BST	BIGCC	8100	8113	–12	9	7615	7623	–8	6	7436	7443	–7	5
<i>End-of-life balance</i>														
Reference	–	–	–319	–363	44	0	–318	–390	72	0	–318	–397	79	0
Recycle Steel substitution	100% ore	50% ore, 50% scrap	–271	–333	62	18	–270	–360	90	18	–270	–367	97	18
<i>Life cycle energy balance</i>														
Reference	–	–	10,716	10,328	388	0	9627	9182	445	0	9326	8870	456	0
Truncated material energy	–	+21%	10,896	10,460	436	48	9818	9330	488	44	9518	9020	497	42
Building lifespan	50 years	100 years	21,102	20,735	366	–22	18,875	18,443	431	–14	18,268	17,824	444	–12
Energy supply	BST	BIGCC	8430	8034	397	9	7996	7545	451	6	7820	7360	460	5
Recycle steel substitution	100% ore	50% ore, 50% scrap	10,764	10,358	406	18	9675	9212	463	18	9374	8900	474	18

greater bioenergy recovery when using wood framing material. These advantages more than outweigh the energy saving benefits of thermal mass. Assuming a 50-year lifespan for the buildings and heat supply from CHP-based district heating, the wood-frame buildings have 4–5% lower life cycle primary energy use than the concrete-frame alternatives, with location in Växjö. The difference decreases to around 2–3% if a 100-year lifespan is assumed, because the energy saving benefits of thermal mass increases proportionally as the lifespan of the buildings get longer. The absolute primary energy benefit of thermal mass is higher when using less energy efficient heating systems but this considerably increases the space heating primary energy use. For example, the space heating primary energy use is more than four times as much when the buildings are electric resistance heated instead of district heated with CHP-based system. The choice of energy supply system is of great importance and should be an integral part of the effort to create a low-energy built environment.

The operation phase dominates the life cycle primary energy use, but less so with energy-efficient buildings and energy-efficient supply systems. As building operation becomes more efficient, the relative importance of the other life cycle phases and the choice of framing material becomes greater.

The production primary energy balance is about twice as much when the buildings are constructed with concrete instead of with wood frames, and the end-of-life primary energy benefit is large when the demolition wood is used for energy purposes. Wood-framed buildings also give greenhouse gas benefits [4–6,45,46]. Experience from Sweden shows that wood multi-storey buildings can be economically competitive to non-wood systems, and issues related to fire and structural performance may be addressed with careful planning and good workmanship techniques [47]. An example is the eight-storey wood-framed apartment building constructed in Växjö. Gustavsson et al. [48] have analyzed the life cycle primary energy use and carbon emissions of this building. The Swedish market share of multi-storey wood-frame buildings has increased from zero to about 15% since the century-old law prohibiting the construction of such building was repealed in 1994 [49].

The trade-offs and synergies associated with different construction alternatives vary and the contribution of all life cycle aspects should be considered to understand the energetic implications of building material choice. Building standards may specify energy performance requirements for design and construction of buildings, and thus can be an important driving force to reduce energy use in buildings. However, existing standards generally focus on the operation energy use and do not reflect the primary energy use during a building's life cycle [26]. In the future, building standards may be developed to consider the production, operation, and end-of-life phases of buildings, including all energy losses in the supply chains to reduce primary energy use and climatic impact of buildings.

We have not studied how the heat storage capacity of building systems interacts with heating systems and how quickly a heating system could respond to a temperature variation and if that has any impact on the heat use. This should be further studied. In our analysis, we accounted for the thermal mass benefit of concrete but did not include the primary energy use for moisture drying in concrete-based material on the construction site. This may underestimate the primary energy use during the production phase of concrete materials.

This analysis shows that a wood-framed building has a lower overall primary energy balance than a concrete-framed building, even if the heat storage capacity associated with thermal mass of the building materials is considered. Considering the thermal mass of buildings in detail does not change the conclusions of Adalberth [3], Gustavsson et al. [4], and Gustavsson and Sathre [5] that wood construction uses less primary energy than concrete construction.

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